Assignment#2:
This exersice is about using the unsteady BEM code to simulate a PI collective pitch controller.

Below rated we want to operate the turbine at its highest C_p that happens near θ_p=0° and a tip speed ratio around 8. To obtain this and ensure that the power does not exceed P_{rated}=10.64MW the generator should be controlled to provide a torque as function of rotational speeds as (compute K and ω_{rated} yourself):

$$M_G(\omega) = \begin{cases} K \cdot \omega^2 & \omega < \omega_{rated} \\ 10.64 \cdot 10^6 / \omega_{rated} & \omega > \omega_{rated} \end{cases}$$

It is be a good idea to make ω_{ref} slightly higher than ω_{rated} to allow for some variation around this reference value in high turbulent wind without going below rated power. Now implement a PI controller in the unsteady BEM code and compute dynamically the rotational speed as

$$I \frac{d\omega}{dt} = M_{aero}(V_o, \theta_{p,}\omega) - M_G(\omega)$$

The inertia moment of the drivetrain is I=$1.6 \cdot 10^8$kgm^2. The gains for this operation are KI=0.64 rad/rad, KP=1.5 rad/(rad/s) and KK=14 deg. Set $\theta_{p,min}$=0° and $\theta_{p,max}$=90°

Remember that the rotational speed is no longer constant so the relative wind becomes

$$\begin{pmatrix} V_{rel,y} \\ V_{rel,z} \end{pmatrix} = \begin{pmatrix} V_{o,y} \\ V_{o,z} \end{pmatrix} + \begin{pmatrix} W_y \\ W_z \end{pmatrix} + \begin{pmatrix} -\omega(t) \cdot r \\ 0 \end{pmatrix}$$

Q#1 Determine the constant K in the generator characteristic, the rated ω_{rated} and the reference rotational speed ω_{ref}. Vary the input wind speed and show that the steady result for a constant wind speed below rated end at the maximum C_p. Also run your unsteady BEM code in time domain to determine the necessary pitch angles $\theta_p(V_o)$ to obtain the rated power of 10.64 MW above rated wind speed and compare with the 10 MW reference report.

Q#2 Finally, also simulate and report results including turbulent wind.

Q#3: Play with ASHES including a controller and compare with your own code